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CSE 250 Circuits & Electronics

Summer 2026

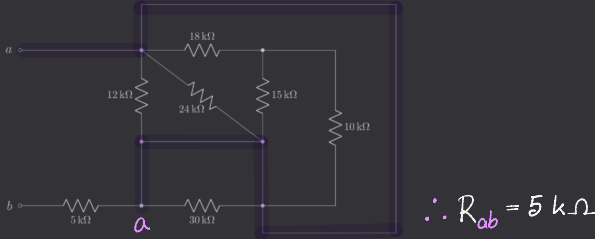
Assignment 01

Section 10A [Tarvin Anjum Aditto]

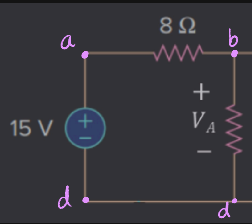
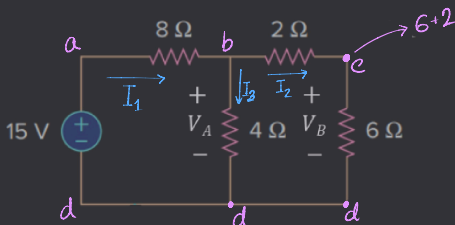
5th July 2026
7:18 p.m.
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Deadline: 6th July 2026
3:29 p.m.

1. Determine R_{ab} , the equivalent resistance between the terminals a and b in the circuit shown below.



2. Find V_A , V_B using voltage division rule. Also find the current through 4 ohm and 2 ohm using current division rule.



VDR

$$6 + 2 \parallel 4 = \frac{8}{3} \Omega \Rightarrow V_A = V_{bd} = \frac{R_1}{\sum R} \cdot V_{ad}$$

$$V_B = V_{cd} = \frac{R_1}{\sum R} \cdot V_A$$

$$= \frac{6}{8} \times \frac{15}{4} = \frac{45}{16} \text{ Volt}$$

$$= \frac{\frac{8}{3}}{\frac{8}{3} + \frac{8}{3}} \times 15 \text{ Volt} = \frac{8^{-1}}{\frac{16}{3}} \times 15 = \frac{15}{4} \text{ Volt}$$

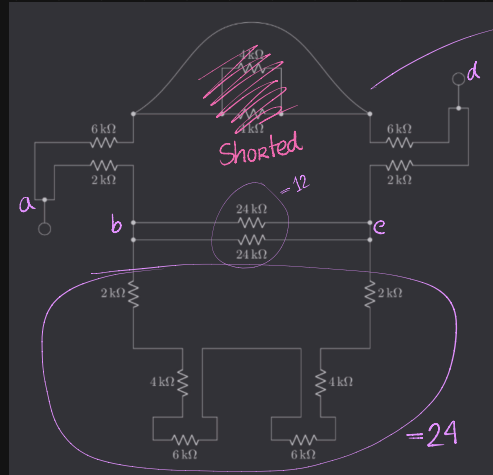
CDR

$$I_1 = \frac{V}{R} = \frac{15}{8}$$

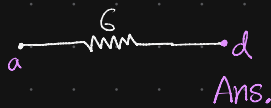
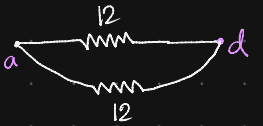
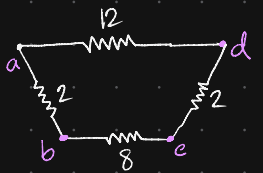
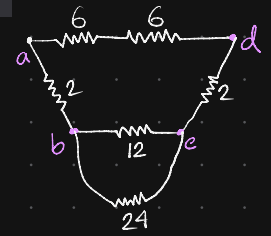
$$I_3 = \text{Across } 4\text{-ohm} = \frac{8^{-1}}{8^{-1} + 4^{-1}} \times I_1 = \frac{5}{8} \text{ Amp}$$

$$I_2 = \text{Across } 2\text{-ohm} = \frac{4^{-1}}{8^{-1} + 4^{-1}} \times I_1 = \frac{10}{8} \text{ Amp}$$

3. Determine the equivalent resistance of the outermost nodes of the resistobot's hands, that is, between the circles in the given figure.

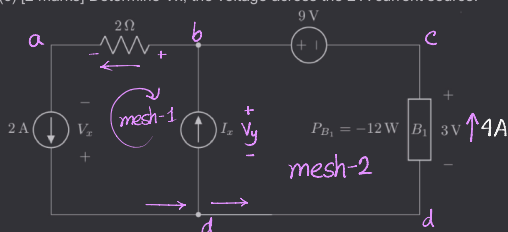


$$R_{eq} = 6 \text{ k}\Omega$$



4. Apply KCL, KVL, and Ohm's Law to answer the following questions—

- (a) [3 marks] Determine the current through the 9V voltage source if the power of the element B1 is -12W.
(b) [3 marks] Determine I_x .
(c) [2 marks] Determine V_x , the voltage across the 2A current source.



a) -ve Power $\Rightarrow$ Supplying electricity

$$P = I \cdot V$$

$$P_{B1} = (-1) \cdot I \cdot 3 \text{ volt} \Rightarrow \therefore I = 4 \text{ Amp}$$

Assume $\uparrow$ Current flow Dir. from node d $\rightarrow$ c

b) KCL @ node-d: $2 = I_x + 4 \Rightarrow I_x = -2 \text{ amp}$

c) KVL @ mesh-1: $V_x + V_{ab} + V_{bd} = 0$
 $V_x - 4 + 12 = 0 \Rightarrow V_x = -8 \text{ Volt}$

$$V_{ab} = -2 \cdot 2 = -4 \text{ V}$$

$$V_{ed} = 3 \text{ V}$$

$$V_{be} = 9 \text{ V}$$

$$\Rightarrow V_{bd} = 12 \text{ V}$$